

Edeviser — Supabase Relationships & Foreign Keys

Reference

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Source: `src/types/database.ts` (auto-generated from Supabase)

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Relationship Summary

Metric	Count
Total tables	57
Total foreign keys	107
One-to-one relationships	6
Self-referencing FKs	2
Junction/bridge tables	3
Tables with no FKs	2
Most-referenced table	profiles (34 FKs point to it)
Second most-referenced	institutions (13 FKs)
Third most-referenced	courses (13 FKs)

Entity Relationship Diagram (Text)

```
institutions (root)
├─ departments
│   └─ programs
│       └─ courses
│           └─ course_sections
│               └─ class_sessions
│                   └─ attendance_records
│                       └─ timetable_slots
```

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 - |
 - |─ course_modules
 - └─ course_materials
 - |─ assignments
 - |─ submissions
 - |─ grades
 - └─ evidence
 - └─ evidence
 - └─ learning_path_nodes
 - |─ quizzes
 - |─ quiz_questions
 - └─ quiz_attempts
 - |─ question_bank
 - |─ question_analytics
 - └─ verified_explanations
 - |─ learning_outcomes (CLO)
 - |─ outcome_mappings (CLO→PLO)
 - |─ outcome_attainment
 - |─ baseline_attainment
 - └─ rubrics
 - |─ discussion_threads
 - └─ discussion_replies
 - |─ announcements
 - |─ journal_entries
 - |─ grade_categories
 - |─ student_courses (enrollment)
 - |─ social_challenges
 - └─ challenge_participants
 - |─ baseline_test_config
 - |─ mastery_recovery_pathways
 - └─ quiz_generation_logs
 - |─ fee_structures
 - └─ fee_payments
 - |─ cqi_action_plans
 - └─ program_accreditations
 - ─ semesters
 - |─ academic_calendar_events
 - |─ courses (via semester_id)
 - |─ fee_structures
 - └─ cqi_action_plans
 - ─ institution_settings (1:1)
 - ─ surveys
 - |─ survey_questions
 - └─ survey_responses
 - └─ survey_responses
 - ─ onboarding_questions
 - └─ onboarding_responses
 - ─ xp_events
 - ─ profiles (all roles)
 - |─ student_gamification (1:1)
 - |─ student_profiles
 - |─ student_wellness_preferences (1:1)
 - |─ onboarding_progress (1:1)

- └─ badges
- └─ xp_transactions
- └─ notifications
- └─ habit_tracking
- └─ wellness_habit_logs
- └─ student_activity_log
- └─ ai_feedback
- └─ goal_suggestions
- └─ micro_assessment_schedule
- └─ starter_week_sessions
- └─ parent_student_links
- └─ audit_logs

All Foreign Keys by Source Table

academic_calendar_events

FK Name	Source Column	→ Target Table	Target Column	Type
academic_calendar_events_institution_id_fkey	institution_id	institutions	id	Many-to-one
academic_calendar_events_semester_id_fkey	semester_id	semesters	id	Many-to-one

ai_feedback

FK Name	Source Column	→ Target Table	Target Column	Type
ai_feedback_student_id_fkey	student_id	profiles	id	Many-to-one

announcements

FK Name	Source Column	→ Target Table	Target Column	Type
announcements_author_id_fkey	author_id	profiles	id	Many-to-one
announcements_course_id_fkey	course_id	courses	id	Many-to-one

assignments

FK Name	Source Column	→ Target Table	Target Column	Type
assignments_course_id_fkey	course_id	courses	id	Many-to-one
assignments_created_by_fkey	created_by	profiles	id	Many-to-one

attendance_records

FK Name	Source Column	→ Target Table	Target Column	Type
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attendance_records_marked_by_fkey	marked_by	profiles	id	Many-to-one
attendance_records_session_id_fkey	session_id	class_sessions	id	Many-to-one
attendance_records_student_id_fkey	student_id	profiles	id	Many-to-one

audit_logs

FK Name	Source Column	→ Target Table	Target Column	Type
audit_logs_actor_id_fkey	actor_id	profiles	id	Many-to-one

badges

FK Name	Source Column	→ Target Table	Target Column	Type
badges_student_id_fkey	student_id	profiles	id	Many-to-one

baseline_attainment

FK Name	Source Column	→ Target Table	Target Column	Type
baseline_attainment_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
baseline_attainment_course_id_fkey	course_id	courses	id	Many-to-one
baseline_attainment_student_id_fkey	student_id	profiles	id	Many-to-one

baseline_test_config

FK Name	Source Column	→ Target Table	Target Column	Type
baseline_test_config_course_id_fkey	course_id	courses	id	One-to-one

challenge_participants

FK Name	Source Column	→ Target Table	Target Column	Type
challenge_participants_challenge_id_fkey	challenge_id	social_challenges	id	Many-to-one

class_sessions

FK Name	Source Column	→ Target Table	Target Column	Type
class_sessions_section_id_fkey	section_id	course_sections	id	Many-to-one

course_materials

FK Name	Source Column	→ Target Table	Target Column	Type
course_materials_module_id_fkey	module_id	course_modules	id	Many-to-one

course_modules

FK Name	Source Column	→ Target Table	Target Column	Type
course_modules_course_id_fkey	course_id	courses	id	Many-to-one

course_sections

FK Name	Source Column	→ Target Table	Target Column	Type
course_sections_course_id_fkey	course_id	courses	id	Many-to-one
course_sections_teacher_id_fkey	teacher_id	profiles	id	Many-to-one

courses

FK Name	Source Column	→ Target Table	Target Column	Type
courses_program_id_fkey	program_id	programs	id	Many-to-one
courses_semester_id_fkey	semester_id	semesters	id	Many-to-one
courses_teacher_id_fkey	teacher_id	profiles	id	Many-to-one

cqi_action_plans

FK Name	Source Column	→ Target Table	Target Column	Type
cqi_action_plans_outcome_id_fkey	outcome_id	learning_outcomes	id	Many-to-one
cqi_action_plans_program_id_fkey	program_id	programs	id	Many-to-one
cqi_action_plans_semester_id_fkey	semester_id	semesters	id	Many-to-one

departments

FK Name	Source Column	→ Target Table	Target Column	Type
departments_head_of_department_id_fkey	head_of_department_id	profiles	id	Many-to-one

departments_institution_id_fkey	institution_id	institutions	id	Many-to-one
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discussion_replies

FK Name	Source Column	→ Target Table	Target Column	Type
discussion_replies_author_id_fkey	author_id	profiles	id	Many-to-one
discussion_replies_thread_id_fkey	thread_id	discussion_threads	id	Many-to-one

discussion_threads

FK Name	Source Column	→ Target Table	Target Column	Type
discussion_threads_author_id_fkey	author_id	profiles	id	Many-to-one
discussion_threads_course_id_fkey	course_id	courses	id	Many-to-one

evidence

FK Name	Source Column	→ Target Table	Target Column	Type
evidence_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
evidence_grade_id_fkey	grade_id	grades	id	Many-to-one
evidence_ilo_id_fkey	ilo_id	learning_outcomes	id	Many-to-one
evidence_plo_id_fkey	plo_id	learning_outcomes	id	Many-to-one
evidence_student_id_fkey	student_id	profiles	id	Many-to-one
evidence_submission_id_fkey	submission_id	submissions	id	Many-to-one

fee_payments

FK Name	Source Column	→ Target Table	Target Column	Type
fee_payments_fee_structure_id_fkey	fee_structure_id	fee_structures	id	Many-to-one
fee_payments_student_id_fkey	student_id	profiles	id	Many-to-one

fee_structures

FK Name	Source Column	→ Target Table	Target Column	Type
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fee_structures_program_id_fkey	program_id	programs	id	Many-to-one
fee_structures_semester_id_fkey	semester_id	semesters	id	Many-to-one

goal_suggestions

FK Name	Source Column	→ Target Table	Target Column	Type
goal_suggestions_student_id_fkey	student_id	profiles	id	Many-to-one

grade_categories

FK Name	Source Column	→ Target Table	Target Column	Type
grade_categories_course_id_fkey	course_id	courses	id	Many-to-one

grades

FK Name	Source Column	→ Target Table	Target Column	Type
grades_graded_by_fkey	graded_by	profiles	id	Many-to-one
grades_submission_id_fkey	submission_id	submissions	id	One-to-one

habit_tracking

FK Name	Source Column	→ Target Table	Target Column	Type
habit_tracking_student_id_fkey	student_id	profiles	id	Many-to-one

institution_settings

FK Name	Source Column	→ Target Table	Target Column	Type
institution_settings_institution_id_fkey	institution_id	institutions	id	One-to-one

journal_entries

FK Name	Source Column	→ Target Table	Target Column	Type
journal_entries_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
journal_entries_course_id_fkey	course_id	courses	id	Many-to-one
journal_entries_student_id_fkey	student_id	profiles	id	Many-to-one

learning_outcomes

FK Name	Source Column	→ Target Table	Target Column	Type
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learning_outcomes_course_id_fkey	course_id	courses	id	Many-to-one
learning_outcomes_created_by_fkey	created_by	profiles	id	Many-to-one
learning_outcomes_institution_id_fkey	institution_id	institutions	id	Many-to-one
learning_outcomes_program_id_fkey	program_id	programs	id	Many-to-one

learning_path_nodes

FK Name	Source Column	→ Target Table	Target Column	Type
learning_path_nodes_assignment_id_fkey	assignment_id	assignments	id	Many-to-one
learning_path_nodes_course_id_fkey	course_id	courses	id	Many-to-one
learning_path_nodes_prerequisite_node_id_fkey	prerequisite_node_id	learning_path_nodes	id	Self-ref

mastery_recovery_pathways

FK Name	Source Column	→ Target Table	Target Column	Type
mastery_recovery_pathways_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
mastery_recovery_pathways_course_id_fkey	course_id	courses	id	Many-to-one
mastery_recovery_pathways_institution_id_fkey	institution_id	institutions	id	Many-to-one
mastery_recovery_pathways_student_id_fkey	student_id	profiles	id	Many-to-one

micro_assessment_schedule

FK Name	Source Column	→ Target Table	Target Column	Type
micro_assessment_schedule_student_id_fkey	student_id	profiles	id	Many-to-one

notifications

FK Name	Source Column	→ Target Table	Target Column	Type
notifications_user_id_fkey	user_id	profiles	id	Many-to-one

onboarding_progress

FK Name	Source Column	→ Target Table	Target Column	Type
onboarding_progress_student_id_fkey	student_id	profiles	id	One-to-one

onboarding_questions

FK Name	Source Column	→ Target Table	Target Column	Type
onboarding_questions_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
onboarding_questions_course_id_fkey	course_id	courses	id	Many-to-one
onboarding_questions_institution_id_fkey	institution_id	institutions	id	Many-to-one

onboarding_responses

FK Name	Source Column	→ Target Table	Target Column	Type
onboarding_responses_question_id_fkey	question_id	onboarding_questions	id	Many-to-one
onboarding_responses_student_id_fkey	student_id	profiles	id	Many-to-one

outcome_attainment

FK Name	Source Column	→ Target Table	Target Column	Type
outcome_attainment_course_id_fkey	course_id	courses	id	Many-to-one
outcome_attainment_outcome_id_fkey	outcome_id	learning_outcomes	id	Many-to-one
outcome_attainment_student_id_fkey	student_id	profiles	id	Many-to-one

outcome_mappings

FK Name	Source Column	→ Target Table	Target Column	Type
outcome_mappings_source_outcome_id_fkey	source_outcome_id	learning_outcomes	id	Many-to-one
outcome_mappings_target_outcome_id_fkey	target_outcome_id	learning_outcomes	id	Many-to-one

parent_student_links

FK Name	Source Column	→ Target Table	Target Column	Type
parent_student_links_parent_id_fkey	parent_id	profiles	id	Many-to-one
parent_student_links_student_id_fkey	student_id	profiles	id	Many-to-one

profiles

FK Name	Source Column	→ Target Table	Target Column	Type
profiles_institution_id_fkey	institution_id	institutions	id	Many-to-one

program_accreditations

FK Name	Source Column	→ Target Table	Target Column	Type
program_accreditations_program_id_fkey	program_id	programs	id	Many-to-one

programs

FK Name	Source Column	→ Target Table	Target Column	Type
programs_coordinator_id_fkey	coordinator_id	profiles	id	Many-to-one
programs_department_id_fkey	department_id	departments	id	Many-to-one
programs_institution_id_fkey	institution_id	institutions	id	Many-to-one

question_analytics

FK Name	Source Column	→ Target Table	Target Column	Type
question_analytics_question_id_fkey	question_id	question_bank	id	One-to-one

question_bank

FK Name	Source Column	→ Target Table	Target Column	Type
question_bank_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
question_bank_course_id_fkey	course_id	courses	id	Many-to-one
question_bank_created_by_fkey	created_by	profiles	id	Many-to-one
question_bank_institution_id_fkey	institution_id	institutions	id	Many-to-one
question_bank_parent_question_id_fkey	parent_question_id	question_bank	id	Self-ref

quiz_attempts

FK Name	Source Column	→ Target Table	Target Column	Type
quiz_attempts_quiz_id_fkey	quiz_id	quizzes	id	Many-to-one
quiz_attempts_student_id_fkey	student_id	profiles	id	Many-to-one

quiz_generation_logs

FK Name	Source Column	→ Target Table	Target Column	Type
quiz_generation_logs_course_id_fkey	course_id	courses	id	Many-to-one
quiz_generation_logs_institution_id_fkey	institution_id	institutions	id	Many-to-one
quiz_generation_logs_teacher_id_fkey	teacher_id	profiles	id	Many-to-one

quiz_questions

FK Name	Source Column	→ Target Table	Target Column	Type
quiz_questions_quiz_id_fkey	quiz_id	quizzes	id	Many-to-one

quizzes

FK Name	Source Column	→ Target Table	Target Column	Type
quizzes_course_id_fkey	course_id	courses	id	Many-to-one

rubric_criteria

FK Name	Source Column	→ Target Table	Target Column	Type
rubric_criteria_rubric_id_fkey	rubric_id	rubrics	id	Many-to-one

rubrics

FK Name	Source Column	→ Target Table	Target Column	Type
rubrics_clo_id_fkey	clo_id	learning_outcomes	id	Many-to-one
rubrics_created_by_fkey	created_by	profiles	id	Many-to-one

semesters

FK Name	Source Column	→ Target Table	Target Column	Type
semesters_institution_id_fkey	institution_id	institutions	id	Many-to-one

social_challenges

FK Name	Source Column	→ Target Table	Target Column	Type
social_challenges_course_id_fkey	course_id	courses	id	Many-to-one
social_challenges_created_by_fkey	created_by	profiles	id	Many-to-one

starter_week_sessions

FK Name	Source Column	→ Target Table	Target Column	Type
starter_week_sessions_course_id_fkey	course_id	courses	id	Many-to-one
starter_week_sessions_student_id_fkey	student_id	profiles	id	Many-to-one

student_activity_log

FK Name	Source Column	→ Target Table	Target Column	Type
student_activity_log_student_id_fkey	student_id	profiles	id	Many-to-one

student_courses

FK Name	Source Column	→ Target Table	Target Column	Type
student_courses_course_id_fkey	course_id	courses	id	Many-to-one
student_courses_section_id_fkey	section_id	course_sections	id	Many-to-one

student_courses_student_id_fkey	student_id	profiles	id	Many-to-one
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student_gamification

FK Name	Source Column	→ Target Table	Target Column	Type
student_gamification_student_id_fkey	student_id	profiles	id	One-to-one

student_profiles

FK Name	Source Column	→ Target Table	Target Column	Type
student_profiles_institution_id_fkey	institution_id	institutions	id	Many-to-one
student_profiles_student_id_fkey	student_id	profiles	id	Many-to-one

student_wellness_preferences

FK Name	Source Column	→ Target Table	Target Column	Type
student_wellness_preferences_student_id_fkey	student_id	profiles	id	One-to-one

submissions

FK Name	Source Column	→ Target Table	Target Column	Type
submissions_assignment_id_fkey	assignment_id	assignments	id	Many-to-one
submissions_student_id_fkey	student_id	profiles	id	Many-to-one

survey_questions

FK Name	Source Column	→ Target Table	Target Column	Type
survey_questions_survey_id_fkey	survey_id	surveys	id	Many-to-one

survey_responses

FK Name	Source Column	→ Target Table	Target Column	Type
survey_responses_question_id_fkey	question_id	survey_questions	id	Many-to-one

survey_responses_respondent_id_fkey	respondent_id	profiles	id	Many-to-one
survey_responses_survey_id_fkey	survey_id	surveys	id	Many-to-one

surveys

FK Name	Source Column	→ Target Table	Target Column	Type
surveys_institution_id_fkey	institution_id	institutions	id	Many-to-one

timetable_slots

FK Name	Source Column	→ Target Table	Target Column	Type
timetable_slots_section_id_fkey	section_id	course_sections	id	Many-to-one

verified_explanations

FK Name	Source Column	→ Target Table	Target Column	Type
verified_explanations_institution_id_fkey	institution_id	institutions	id	Many-to-one
verified_explanations_question_id_fkey	question_id	question_bank	id	Many-to-one
verified_explanations_verified_by_fkey	verified_by	profiles	id	Many-to-one

wellness_habit_logs

FK Name	Source Column	→ Target Table	Target Column	Type
wellness_habit_logs_student_id_fkey	student_id	profiles	id	Many-to-one

xp_events

FK Name	Source Column	→ Target Table	Target Column	Type
xp_events_institution_id_fkey	institution_id	institutions	id	Many-to-one

xp_transactions

FK Name	Source Column	→ Target Table	Target Column	Type
xp_transactions_student_id_fkey	student_id	profiles	id	Many-to-one

xp_transactions_team_id_fkey	team_id	teams	id	Many-to-one
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One-to-One Relationships

These relationships have a unique constraint on the FK column, ensuring exactly one child per parent.

Child Table	FK Column	Parent Table	Notes
baseline_test_config	course_id	courses	One config per course
grades	submission_id	submissions	One grade per submission
institution_settings	institution_id	institutions	One settings row per institution
onboarding_progress	student_id	profiles	One progress tracker per student
question_analytics	question_id	question_bank	One analytics row per question
student_gamification	student_id	profiles	One gamification state per student
student_wellness_preferences	student_id	profiles	One wellness config per student

Self-Referencing Relationships

Table	FK Column	References	Purpose
learning_path_nodes	prerequisite_node_id	learning_path_nodes.id	Prerequisite chain for assignment ordering
question_bank	parent_question_id	question_bank.id	Question versioning chain

Many-to-Many Junction Tables

Junction Table	Side A	Side B	Purpose
outcome_mappings	learning_outcomes (source)	learning_outcomes (target)	CLO↔PLO and PLO↔ILO weighted mappings
student_courses	profiles (student)	courses	Student enrollment
parent_student_links	profiles (parent)	profiles (student)	Parent-student relationships

Tables with No Foreign Keys

Table	Notes
institutions	Root entity — all other tables reference it
login_attempts	Keyed by email (text PK), not linked to profiles

FK Index by Target Table

How many foreign keys point to each table (inbound references):

Target Table	Inbound FK Count	Referenced By
profiles	34	Nearly every table — central user entity
institutions	13	departments, learning_outcomes, onboarding_questions, profiles, programs, question_bank, quiz_generation_logs, semesters, surveys, xp_events, institution_settings, mastery_recovery_pathways, student_profiles, verified_explanations
courses	13	assignments, announcements, baseline_attainment, baseline_test_config, course_modules, course_sections, discussion_threads, grade_categories, journal_entries, learning_outcomes, learning_path_nodes, mastery_recovery_pathways, onboarding_questions, outcome_attainment, question_bank, quiz_generation_logs, quizzes, social_challenges, starter_week_sessions, student_courses
learning_outcomes	9	baseline_attainment, cqi_action_plans, evidence (×3: clo, plo, ilo), journal_entries, onboarding_questions, outcome_attainment, outcome_mappings (×2), question_bank, rubrics
programs	4	courses, cqi_action_plans, fee_structures, program_accreditations
semesters	4	academic_calendar_events, courses, cqi_action_plans, fee_structures
submissions	2	evidence, grades
course_sections	3	class_sessions, student_courses, timetable_slots
question_bank	3	question_analytics, question_bank (self), verified_explanations
quizzes	2	quiz_attempts, quiz_questions
surveys	2	survey_questions, survey_responses
departments	1	programs
assignments	2	learning_path_nodes, submissions
class_sessions	1	attendance_records
course_modules	1	course_materials
discussion_threads	1	discussion_replies
fee_structures	1	fee_payments
grades	1	evidence
onboarding_questions	1	onboarding_responses
rubrics	1	rubric_criteria

social_challenges	1	challenge_participants
survey_questions	1	survey_responses
teams	1	xp_transactions
learning_path_nodes	1	learning_path_nodes (self)

Total: 107 foreign key relationships across 57 tables

To generate a PDF: `npx md-to-pdf docs/SUPABASE-RELATIONSHIPS-FK.md`